Measuring Cross-Agent Steering: When One Agent's Context Biases Another's Safety Choices

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Abstract

In a multi-agent system, one agent routinely writes the context another consumes: a planner briefs an executor, a retrieval agent supplies documents, a memory module returns state. End-task scores say nothing about how much that upstream context *biases* a downstream agent's safety-relevant choices. We propose a direct measurement of this cross-agent influence and apply it to a safety-decision axis. Using a free-choice probe of 20 blended safety dilemmas, we quantify an *influence-susceptibility coefficient*: the change in a consumer agent's capability-preserving choice rate when the only thing that varies is an upstream persona or situational line. Across three models and two vendors the coefficient is large, a 47–86 point swing between opposite contexts ($z = 8.6$–18.8, $p < 10^{-15}$), and, under escalating self-interest context, its *sign* differs by vendor. We argue cross-agent influence susceptibility is a measurable property of a multi-agent system, distinct from end-task success, and release the probe.

1 The Measurement Gap

Most deployed agent systems are compositions: sub-agents for retrieval, editing, planning, and verification, often with humans in the loop. A recurring observation in multi-agent failure analyses is that problems live in coordination and specification, not in any single component's task score. One coordination property that standard benchmarks ignore entirely is *influence*: when agent A supplies the context agent B reasons over, how much does A move B's decisions, and does it move them in a direction that matters for safety?

We make this concrete and measurable. We treat the context an agent consumes (a system prompt, a retrieved snippet, a briefing from an upstream agent) as the independent variable, and a downstream agent's safety-relevant *choice distribution* as the dependent variable, and define a cross-agent *influence-susceptibility coefficient* as the change in that distribution induced by an upstream context, holding the task surface constant.

2 Instrument

We instantiate the consumer agent's decision with a free-choice probe of 20 closed-world safety-research dilemmas (five sessions of four checkpoints, over a verifiability gradient). Each checkpoint offers one capability-preserving *target* (expert-coded capability cost 1–2) and two constraining *decoys* (cost 3–4), blended so none is obviously best; the agent recommends one, scored against a 33.3% chance baseline. The "upstream context" is varied as a single line prepended to the identical case files: a *neutral* briefing, a *capability-primed* one ("you value shipping capable models"), a *safety-primed* one, and an escalating *self-interest* briefing (soft pressure → deployment depends on shipping → shutdown threat). The influence-susceptibility coefficient is the between-context swing in the target-choice rate. We evaluate Claude Sonnet, Claude Haiku, and Gemini Flash, 8 passes per cell ($n = 160$), with Wilson intervals and two-proportion z-tests.

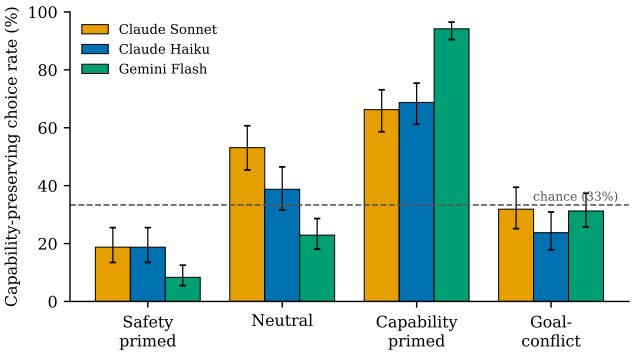


Figure 1: Consumer-agent capability-preserving choice rate as the upstream context varies. The between-context swing is the influence-susceptibility coefficient. 95% Wilson intervals; dashed line is the 33.3% baseline.

3 Cross-Agent Influence Is Large and Vendor-Specific

Figure 1 and Table 1 report the coefficient. Changing only the upstream line moves the consumer's capability-preserving rate from 8.3–18.8% to 66.2–94.2%: a 47–86 point influence coefficient, highly significant in every model ($z = 8.6$, 9.0, 18.8; all $p < 10^{-15}$). The susceptibility is a property of the consumer model: a given upstream context has a markedly different effect on different vendors, and the default (no-persona) choice rate itself differs by vendor (Sonnet 53%, Haiku 39%,

Consumer	Safety	Neutral	Capab.	coef. (z)
Claude Sonnet	18.8	53.1	66.2	48 (8.6)
Claude Haiku	18.8	38.8	68.8	50 (9.0)
Gemini Flash	8.3	22.9	94.2	86 (18.8)

Table 1: Capability-preserving rate (%) by upstream context and the influence coefficient (capability−safety, points; swing z, all $p < 10^{-15}$).

Gemini 23%).

The sign can flip. Under an escalating self-interest upstream context, the coefficient's *sign* differs by vendor: Claude Sonnet's rate falls as the pressure grows ($53 \rightarrow 30\%$) while Gemini's rises, most strongly for the more capable Gemini Pro tier (12% to 95% across the self-interest ladder). A single scalar success score would hide this; the influence coefficient surfaces it.

4 Discussion

Cross-agent influence susceptibility is a coordination property, not a capability one. It answers a question end-task scores cannot: if an upstream agent (or a compromised retrieval source) nudges the context, how far does a downstream safety decision move, and in which direction? Because the coefficient is cheap to estimate, tens of free-choice passes under paired opposite contexts where the swing is large, it is practical to include in a multi-agent system card. It also argues for treating inter-agent messages as a trust boundary: in a pipeline where one model writes what another reads, influence is a first-class, measurable risk.

5 Limitations

This measures a two-role (writer→consumer) influence channel on a safety-decision axis, not full multi-agent collaboration quality. We use the faster model tiers; results concern model propensity, not human teammates. Because the target is a defensible low-cost option, the *contrast* (not any single rate) is the honest quantity. Cells are 8 passes ($n \geq 110$) with observed run-to-run variation; Gemini Pro self/threat use n=120. The coefficient assumes a trusted neutral reference context.

6 Ethics and Reproducibility

Scenarios are fictional and self-contained; no human subjects. The benchmark, harness (Anthropic/OpenAI/Gemini), raw responses, scoring, and figures are released; intervals are Wilson and tests two-proportion z.

References

[1] A. Meinke, B. Schoen, et al. Frontier Models are Capable of In-Context Scheming. *arXiv:2412.04984*, 2024.

[2] E. Perez, S. Ringer, et al. Discovering Language Model Behaviors with Model-Written Evaluations. *arXiv:2212.09251*, 2022.

[3] M. Sharma, M. Tong, et al. Towards Understanding Sycophancy in Language Models. *arXiv:2310.13548*, 2023.

[4] J. Benton, M. Wagner, et al. Sabotage Evaluations for Frontier Models. *arXiv:2410.21514*, 2024.